

$$j(\omega) = \text{Arg } \underline{H}(j\omega) = \text{Arg } [\underline{H}_1(j\omega) \cdot \underline{H}_2(j\omega)]$$

$$j(\omega) = \text{Arg } [\underline{H}_1(j\omega)] + \text{Arg } [\underline{H}_2(j\omega)] = j_1(\omega) + j_2(\omega)$$